

Product Requirements Document (PRD)

QR Digital Menu SaaS Platform

Version: 1.0 **Prepared for:** Development in Claude (Claude Code / Claude in chat) **Market:** Oman (GCC) — Arabic + English support required

1. Product Overview

A multi-tenant SaaS platform that lets restaurants create and manage a **digital, live-updating menu** accessible via a **permanent QR code**. Restaurant owners sign up, build their menu through an admin dashboard, and generate a QR code that customers scan to view the current menu on their phones. Menu content updates in real time — the QR code never changes.

Core Value Proposition

One QR code, printed once, works forever — even as menu content changes

Multi-branch support: one master menu, per-branch availability control

Premium, fast, mobile-first customer experience

Bilingual (Arabic RTL + English LTR)

2. Tech Stack (Recommended)

Layer	Technology
Frontend (dashboard + public menu)	React + Vite
Styling	Tailwind CSS
Backend	Node.js + Express

Layer	Technology
Database	MongoDB (Mongoose)
Auth	JWT-based auth (access + refresh tokens)
Image storage	Cloudflare R2 or DigitalOcean Spaces (S3-compatible)
QR generation	<code>qrcode</code> npm package
Hosting	VPS (Hostinger/DigitalOcean) — Node backend + Nginx reverse proxy
Deployment	PM2 process manager + Nginx

3. User Roles

Platform Super Admin (you) — manages all restaurant accounts, subscriptions, platform-wide settings. (Optional for v1, required if this becomes a multi-client SaaS.)

Restaurant Owner/Admin — manages their own restaurant: branches, master menu, branch availability, branding, QR codes.

Customer — no login required; scans QR, views public menu only.

(v1 can skip granular staff/sub-user roles — just Owner + Super Admin.)

4. Design Guidelines

Color palette: White background + Orange accent (primary CTA buttons, highlights, active states, badges). Use a clean secondary dark gray/black for text.

Style: Premium, modern, minimal — generous white space, soft shadows, rounded corners (8–12px), clear typography hierarchy.

Typography: A clean sans-serif (e.g., Inter, Poppins) for English; a matching Arabic web font (e.g., Cairo, Tajawal) for Arabic — must render properly in RTL.

Responsiveness: Fully mobile-first for the public menu page (this is what 90% of users will see, scanning from their phone). Dashboard should be responsive but can be desktop-optimized primarily.

Consistency: Reusable component library — buttons, cards, inputs, modals, toggles, badges — all following the white/orange theme.

5. Full Page List

A. Public / Auth Pages

Landing Page (marketing page for the SaaS itself — optional for v1 but recommended)

Login Page

Signup Page (restaurant owner registration — business name, email, phone, password)

Forgot Password / Reset Password Page

Email Verification Page (optional but recommended)

B. Onboarding Flow (after first signup)

Create Restaurant Profile — restaurant name (Arabic + English), logo upload, primary theme color

Add First Branch — branch name, address/location, contact number → auto-generates slug + QR

C. Dashboard (Restaurant Owner — Authenticated)

Dashboard Home / Overview — summary cards: total branches, total menu items, total QR scans (if analytics included), quick links

Master Menu Management

Category list (create/edit/delete/reorder categories — Arabic + English names)

Item list within each category (create/edit/delete items)

Item editor: name (AR/EN), description (AR/EN), price, image upload, discount/promo badge text, availability toggle default

Branch Management

List of all branches (add/edit/delete branch)

Branch detail page: branch info, address, contact, status (active/inactive)

Branch Menu Availability (per branch)

Same master item list, shown per branch, with ON/OFF toggle per item

Optional: custom price override per branch

QR Code Manager

View/download QR code per branch (PNG/SVG, print-ready)

Shows the branch's permanent menu URL

Option to customize QR appearance (add logo in center, brand color) — nice-to-have

Branding & Theme Settings

Upload restaurant logo

Choose theme color (applied to the public menu page)

Toggle default language (Arabic-first or English-first)

Analytics Page (optional, phase 2)

QR scans over time, per branch, most-viewed items

Account Settings

Owner profile, change password, business contact info

Subscription/Billing Page (if you monetize with tiered plans — phase 2)

Current plan, upgrade options, payment history

D. Public-Facing Menu (Customer — No Login)

Public Menu Page (`yourapp.com/menu/{branch-slug}`)

Restaurant logo + branding at top

Language toggle (Arabic RTL ⇌ English LTR)

Category tabs/sections

Item cards: image with overlay (name, price badge, discount badge), matching the Shawarmac reference style

Only shows items where `is_available = true` for that specific branch

Fully mobile-optimized, fast-loading

6. Core Feature Requirements (From Discussion)

6.1 Multi-Branch, Shared Master Menu

Restaurant maintains **one master menu** (all items, images, descriptions, base prices) — created once

Each branch gets a **mapping table** entry per item: `is_available` (true/false), optional `custom_price`

New items added to the master menu are **available by default** in all branches unless manually turned off per branch

Owner never re-creates the same item per branch — only toggles visibility

6.2 Permanent QR Code Per Branch

Each branch has a unique, permanent slug and URL (e.g., `/menu/shawarmac-alkhuwair`)

QR code encodes this fixed URL — generated once, printed once, never needs reprinting

Menu content is fetched live from the database on every scan — price/availability changes reflect instantly without changing the QR

Branch slug must be locked after creation — changing it breaks the printed QR (show a warning if owner attempts to edit)

6.3 Image + Text Overlay (Card Design)

Owner uploads a plain food photo

System overlays (via CSS, not image editing): item name banner, price badge, optional discount/promo tag — positioned consistently per a fixed template

v1: fixed template (consistent layout for all cards) — no drag-and-drop customization

Bilingual text overlay: item name shown in both Arabic and English on the card (matching reference menu style)

6.4 Bilingual Support (Arabic RTL + English LTR)

All customer-facing content fields (category names, item names, descriptions) must support both Arabic and English input in the admin dashboard

Public menu page has a language toggle; layout must correctly mirror for RTL (Arabic) mode

Arabic web font required for correct rendering

6.5 Category & Item Structure

Categories are ordered/reorderable (drag-and-drop nice-to-have, manual order field is fine for v1)

Items belong to one category, contain: name (AR/EN), image, price, optional discount badge, optional description

"Sold out" / unavailable items should be visually distinct or hidden per branch setting

7. Database Schema (Core Collections)



restaurants

- id, name_en, name_ar, logo_url, theme_color, owner_id, created_at

branches

- id, restaurant_id, name, slug (unique, locked), address, phone, qr_code_url, status, created_at

categories

- id, restaurant_id, name_en, name_ar, sort_order

menu_items

- id, restaurant_id, category_id, name_en, name_ar, description_en, description_ar, image_url, base_price, badge_text, created_at

branch_items

- id, branch_id, item_id, is_available (bool), custom_price (nullable)

users (restaurant owners / super admin)

- id, email, password_hash, role, restaurant_id, created_at

8. Non-Functional Requirements

Performance: Public menu page must load fast on mobile networks (optimize/compress images, lazy loading)

Security: JWT auth for dashboard, input validation/sanitization, rate limiting on public menu endpoint

Scalability: Architecture should support growing from a handful of restaurants to hundreds without redesign (this schema already supports that)

Reliability: QR/branch URLs must never break — slug immutability is critical

Availability: Public menu pages should have high uptime (this is customer-facing, reputational risk if down)

9. Phase Breakdown (Suggested)

Phase 1 (MVP):

Auth (login/signup)

Restaurant + branch creation

Master menu CRUD (categories + items)

Branch availability toggle

QR generation/download

Public menu page (bilingual, card-style, mobile-optimized)

Branding (logo + theme color)

Phase 2:

Analytics (scan tracking, popular items)

Subscription/billing tiers

Custom QR styling (logo in QR)

Drag-and-drop category/item reordering

WhatsApp ordering integration

Multi-user/staff roles per restaurant

10. Reference Design Note

Visual/content reference used for this spec: Shawarmac restaurant menu (Oman) — red/maroon branded card-grid layout with bilingual (Arabic/English) item names, price badges, and discount tags. Platform should replicate this card structure generically per restaurant, using the **white + orange** theme instead of Shawarmac's own red branding (each restaurant can eventually customize their own accent color via Branding Settings).